HLA-ProtBERT: Multi-Encoder HLA Analysis

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In Brief

- Production-ready pipeline with 5 encoders (ProtBERT, ESM-2, ProtT5, Ankh base/large)
- Caches embeddings/plots for 26k IMGT/HLA alleles (latest snapshot)
- Class I focus: 17,279 A/B/C alleles clustered across PCA, t-SNE, UMAP
- Resumable CLI scripts, GPU-aware, SSL bypass + HF token support

Benefits

- **Mgmt:** Efficient analysis pipeline
- **Clinical:** Better matching insights
- **Tech:** Novel functional view

Overview

1. HLA Analysis Challenge
2. AI-Based Approach
3. Key Findings
4. Applications

Content for:

- Management
- Clinical Scientists
- Bioinformaticians

HLA Complexity

- Thousands of alleles with critical immune function
- Sequence similarity is not equivalent to functional similarity
- Growing database makes manual analysis impossible

Clinical: Better transplant matching
Mgmt: Efficiency gains
Tech: Novel computational approach

Key Takeaway

HLA complexity requires advanced methods beyond simple sequence comparison.

HLA-ProtBERT: Multi-Encoder Pipeline

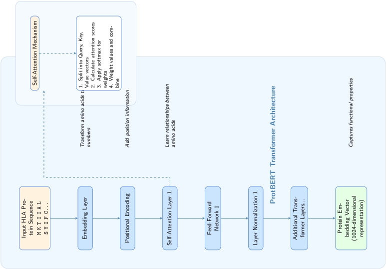
- Encoders: ProtBERT (768), ESM-2 (1280), ProtT5 (1024), Ankh base (768), Ankh large (1536)
- Shared encoder base: caching, locus filtering, SSL toggle, optional HF token
- GPU auto-detection; batch encoding to handle full IMGT/HLA catalog
- Pipeline script orchestrates downloads, embedding generation, and locus plots

Clinical: Finds functionally similar HLAs
Mgmt: Leverages existing AI advances
Tech: Transfer learning from vast protein data

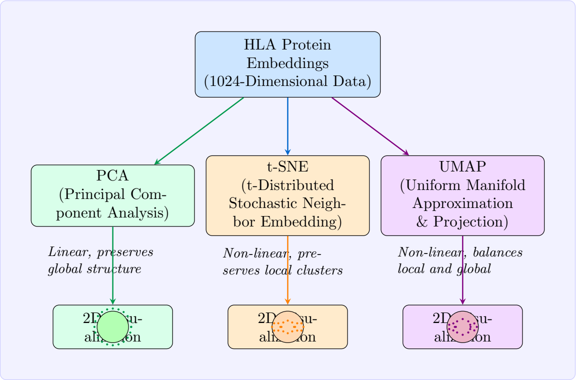
Key Takeaway

ProtBERT understands protein "language" beyond simple sequence similarity.

How It Works



- Processes HLA sequence considering all amino acid relationships
- Multiple layers extract complex patterns
- Output: vector capturing protein properties



Dimensionality Reduction Techniques

PCA
Global overview

t-SNE
Local clusters

UMAP
Balanced view

Encoder 1: ProtBERT

- **Source:** RostLab (2020)
- **Architecture:** BERT-Large (Transformers)
- **Scale:** 420 Million parameters, 30 layers
- **Training Data:** UniRef100 (216M protein sequences)
- **Output:** 1024-dimensional embeddings (per residue)

"The 'Swiss Army Knife' of protein language models - trained on the entire known protein universe."

Why we use it

Excellent distinctiveness between HLA loci. Shows strong correlation with peptide binding properties.

Encoder 2: ESM-2

- **Source:** Meta AI (Facebook Research)
- **Architecture:** Modern Transformer with Rotary Embeddings (RoPE)
- **Scale:** 650 Million parameters (in our pipeline)
- **Training Data:** UniRef50 (Evolutionary balanced)
- **Output:** 1280-dimensional embeddings

"Evolutionary Scale Modeling - predicts structure directly from sequence."

Why we use it

Captures subtle evolutionary relationships. The high dimensional output provides rich structural context.

Encoder 3: Ankh

- **Source:** Elnaggar/RostLab (2023)
- **Architecture:** Protein-Specific Transformer
- **Scale:** 50M (Base) to 650M (Large) parameters
- **Training Data:** UniRef50
- **Output:** 768 or 1536-dimensional embeddings

"Optimized specifically for the language of proteins, not just adapted from text."

Why we use it

Highly efficient inference. The "Large" model rivals ESM-2 in performance while being faster for production matching.

Vignette: The "Mismatch" That Wasn't

Scenario: A patient needs a transplant. We find a donor with a single mismatch on the HLA-A locus.

Standard Analysis

Donor: A*02:01
Patient: **A*03:01**
Verdict: **MISMATCH**
(Different serological groups)

→

AI Analysis (ProtBERT)

Similarity Score: **0.997** (High)
Verdict: **Functional Match**
Physicochemical profile of the peptide binding groove is 99.7% identical.

Key Takeaway

The AI sees beyond the label: A*02 and A*03 share significant functional properties.

Case Study: Donor-Recipient Analysis

Match Report

Overall Score: 0.9664

High functional compatibility despite allelic differences.

Locus-Level Insights:

- **HLA-A:** High Similarity
  - A*01:01 vs A*24:02 → 0.9966
  - A*02:01 vs A*01:01 → 0.9954
- **HLA-B:** Excellent Match
  - B*07:02 vs B*15:01 → 0.9981
- **HLA-C:** Functional Divergence
  - **C*07:01 vs C*03:04 → 0.8010**

Key Takeaway

The model flags the C-locus mismatch (0.80) as a potential risk factor hidden by the high overall score.

Deep Dive: Similarity Search

Query: Find neighbors for `A*01:01`

1. Exact/Synonymous Matches (Sim = 1.0)

- A*01:01:01:01
- A*01:395
- A*01:251

Validates model precision on known variants.

2. Functional Cross-Locus Links

- **A*01:01 vs B*07:02 → 0.9857**
  "Very similar protein group"
- **A*01:01 vs C*07:01 → 0.7745**
  "Substantially different"

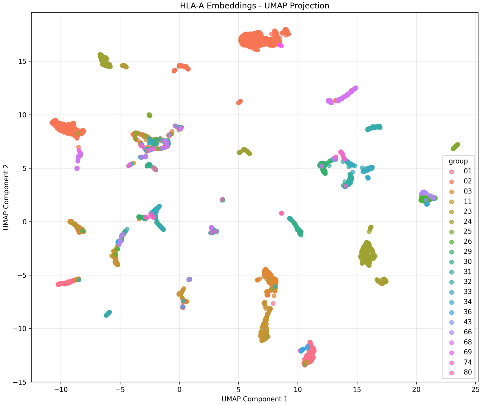
Quantifies biological distance beyond nomenclature.

Analysis Overview

- 17,279 Class I alleles analyzed (ProtBERT embeddings):
  - 5,489 HLA-A
  - 6,584 HLA-B
  - 5,206 HLA-C
- Additional Class II data available (DRB1, DQB1, DPB1, etc.)
- Clear clustering by functional groups across PCA/t-SNE/UMAP

Clinical: Reveals functional similarities
Mgmt: Demonstrates efficiency at scale
Tech: Effective protein embedding

HLA-A Visualization



- Clear separation between major allele families
- A*01, A*02, A*03 groups form discrete clusters

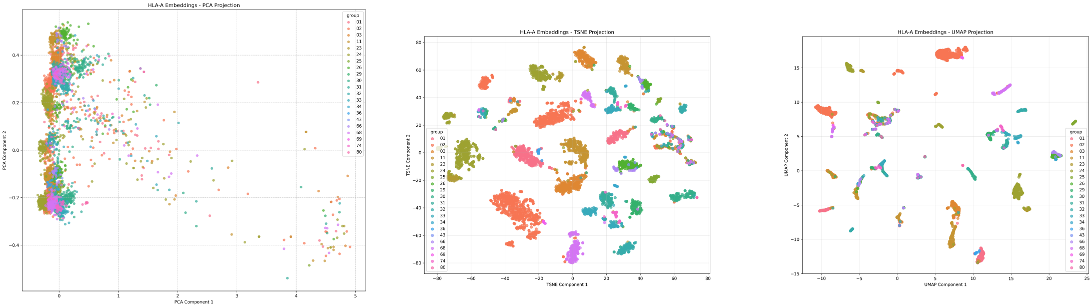
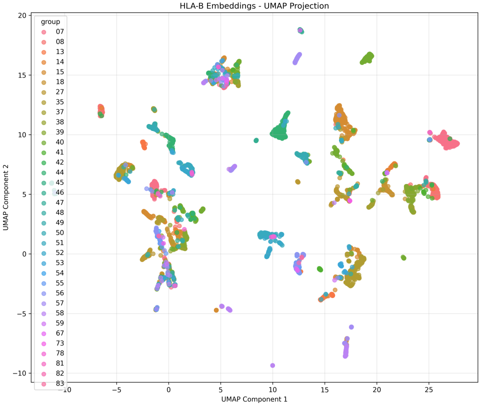


Figure: PCA (left), t-SNE (middle), UMAP (right)

Clinical: Helps identify potential cross-reactive epitopes

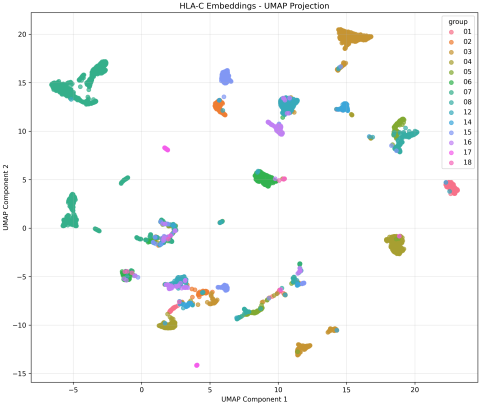
Tech: Different methods reveal different relationship aspects

HLA-B: Most Diverse



- Most diverse locus (6,526 alleles)
- Complex clustering reflects greater polymorphism

HLA-C: Distinct Patterns



- More compact clusters than A and B loci
- Clearer separation between major groups

Current Applications

Clinical

- **Clinical:** Better matching prediction
- **Clinical:** Novel allele classification
- **Clinical:** Cross-reactivity assessment

Research & Technical

- **Tech:** Structure-function modeling
- **Mgmt:** Efficient analysis workflow
- **Tech:** Evolutionary analysis

> Understanding functional relationships between HLA alleles has direct applications in transplantation, disease association, and drug response prediction.

Future Directions

Technical Advancements

- HLA-specific training
- Structural information integration
- Multi-modal models

Data Integration

- Clinical outcome correlation
- Population genetics
- Cross-species analysis

Potential Impact

- **Clinical:** Improved virtual crossmatching
- **Mgmt:** Better matching algorithms
- **Tech:** Novel evolutionary insights

Key Takeaways

1. Multi-encoder embeddings reveal functional HLA relationships
2. 17,279 Class I alleles analyzed from the Dec 2025 IMGT/HLA snapshot
3. Clear functional clustering across PCA, t-SNE, and UMAP
4. Resumable pipeline caches embeddings for ProtBERT, ESM-2, ProtT5, Ankh
5. Applications span clinical, research, and production workflows

Clinical: Explore matching prediction applications

Mgmt: Consider workflow integration

Tech: Explore additional models

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Technical Details

Model Specifications

- ProtBERT: 420M params, 768-dim
- ESM-2 (650M): 1280-dim
- ProtT5 XL: 1.3B params, 1024-dim
- Ankh Base/Large: 50M/650M, 768/1536-dim
- Ensemble-ready; embeddings cached under `data/embeddings/`

Dimensionality Reduction Parameters

- **UMAP:** n_neighbors=15, min_dist=0.1
- **t-SNE:** perplexity=30, learning_rate=200
- **PCA:** n_components=2

Implementation

- `run_complete_pipeline_all_encoders.sh` orchestrates download & encoding
- Hugging Face downloads with optional HF token; `--disable-ssl-verify` flag
- GPU auto-detection; batch encoding defaults to 8
- Caching + resumable runs for embeddings and locus plots
- Locus analysis via `scripts/run_locus_analysis.py` (PCA/t-SNE/UMAP)